Numpy, Matplotlib, Pandas exercises

Irene Siragusa, PhD

CHILAB
Università degli Studi di Palermo

E1 – Standardization & Normalization

- Given a matrix D of 30 datapoints, where each of them is characterized by 9 features:
  - Define a function to print the matrix
  - Transpose the matrix;
  - Calculate min, max and mean values;
  - Standardize and normalize the data in the matrix.

E2 – Function plotting

- Plot in the same graph a linear, a quadratic and a cubic function.
- Add title and legend and differentiate each line per color and linestyle

E3 - Subplots

- Create a simple data set of items in a supermarket, where to each item is associated the corresponding quantity.
- Create a plot made up of three subplots in which data are represend with bars, scatter plot and classic line-plot, with legend and proper personalization of labels and colors.

E4 – Syntetic data set

- What is the scope of this code?

```
import numpy as np
import pandas as pd

rng = np.random.default_rng(42)
n = 200
df = pd.DataFrame({
    'id': np.arange(1, n+1),
    'department': rng.choice(['HR','IT','Sales','R&D'], size=n),
    'age': rng.integers(22, 65, size=n),
    'height': rng.normal(1.70, 0.09, n).round(2),
    'weight': rng.normal(72, 12, n).round(1),
})
df['salary'] = (22000 + df['age']*450 + rng.normal(0, 4000, n)).round(0)
```

E4 – Syntetic data set

- Execute the previous code and build the data set
- Get basics statistics of the data set
- Plot useful graphs of the data set

- Set about 8% of salary and 5% of age to NaN, using a random boolean mask.
- Append 5 duplicated rows with pd.concat.
- Save with to_csv(..., sep=';', decimal=',').

E5 – Selection and filtering

- Open the previous saved csv file
- Find and print:
  - employees older than 40 in IT
  - name/department/salary of the 5 oldest employees
  - departments in ['HR','IT'] with age between 30 and 50
  - the cell at row label 10 and column salary

E6 – Feature engineering

- Add bmi feature, calculated as
  - bmi = weight / height**2
- Implement a strategy to handle missing data
- Create age_int feature where age is divided in equi-width bins
- Create salary_int feature where salary is divided in equi-depth bins
- Group by ['department', 'age_int'] and show the mean salary

E7 – Data preparation pipeline

- Download the data set from Kaggle
  - https://www.kaggle.com/datasets/himelsarder/retail-product-dataset-with-missing-values

- Analyze the data set (statistics, plots, …)
- Implement a strategy to infer missing data
- Discretize and binarize data types